Probability Lecture Notes 1: Basics

1. Meaning of Probability

Probability is the branch of mathematics that measures how likely an event is to happen. A probability is always between 0 and 1.

A probability of 0 means an event is impossible. A probability of 1 means an event is certain. A probability such as 0.5 means the event has a 50% chance of occurring.

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0 <= P(A) <= 1
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2. Important Terms

Experiment: An action that produces a result, such as tossing a coin or rolling a die.

Outcome: A single possible result of an experiment.

Sample space: The set of all possible outcomes.

Event: One outcome or a group of outcomes that we are interested in.

3. Sample Space Examples

For one coin toss, the sample space is: S = {Heads, Tails}.

For one six-sided die, the sample space is: S = {1, 2, 3, 4, 5, 6}.

If the event is 'rolling an even number', then E = {2, 4, 6}.

4. Basic Probability Formula

When all outcomes are equally likely, probability is calculated using:

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P(A) = Number of favourable outcomes / Total number of possible
outcomes
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Example: The probability of rolling a 5 on a fair six-sided die is 1/6 because there is one favourable outcome and six possible outcomes.

5. Worked Example

Question: What is the probability of rolling an even number on a fair die?

Even outcomes are 2, 4 and 6. Therefore there are 3 favourable outcomes out of 6 possible outcomes.

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P(even) = 3/6 = 1/2 = 0.5 = 50%
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6. Theoretical and Experimental Probability

Theoretical probability is based on mathematical reasoning. For a fair coin, P(Heads) = 1/2.

Experimental probability is based on actual results from repeated trials.

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Experimental P(A) = Number of times A occurs / Total number of trials
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Example: If a coin is tossed 100 times and lands on heads 47 times, the experimental probability of heads is 47/100 = 0.47.

7. Quick Review

1. What is the sample space for one die roll?

2. What is the probability of rolling a number greater than 4?

3. What is the difference between theoretical and experimental probability?